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Chemoattractant-Elicited Increases in Myosin Phosphorylation in Dictyostelium

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Summary

Cyclic AMP stimulation of chemotactically competent Dictyostelium amebas labeled with [^{32}P]orthophosphate transiently increases phosphorylation in the heavy chain and the 18,000 dalton light chain of myosin. Immediately before the increase, heavy chain phosphorylation transiently decreases. These phosphorylation changes also occur when cAMP-induced activation of adenylate cyclase is blocked by pretreatment of amebas with caffeine. The time course of these phosphorylation responses correlates with the shape changes induced in amebas exposed to a temporal increase in cAMP concentration. The dose dependence of the phosphorylation responses is the same as that previously determined for chemotaxis. The phosphorylation responses exhibit adaptation properties in common with those of the shape change response and chemotaxis. Increases in the rate of myosin heavy chain and light chain phosphorylation can be observed in vitro by stimulating unlabeled amebas with cAMP and then lysing the cells into a γ-[^{32}P]ATP-containing reaction mixture.

Introduction

Actins and myosins similar to those in muscle are found in all nonmuscle eukaryotic cells and are thought to be involved in numerous functions including ameboid movement, cytoplasmic streaming, changes in cell shape, cell division, phagocytosis, and secretory processes (Clarke and Spudich, 1977; Korn, 1978).

We have focused on Dictyostelium discoideum as a system for studying the roles of actin and myosin in cell motility. Dictyostelium myosin has been characterized and shown to resemble muscle myosin in many of its properties (Clarke and Spudich, 1974). It is composed of two heavy chains of 210,000 daltons and two each of two classes of light chains of 18,000 and 16,000 daltons. When purified from ^{32}P-labeled cells, it is phosphorylated in vivo on both the heavy chain and the 18,000 dalton light chain (Kuczmarski and Spudich, 1980). Increased phosphorylation of the heavy chain by a kinase from growing cells was found to inhibit assembly of bipolar thick filaments at physiological salt concentrations (about 50 mM KCl) and to inhibit actin-activated ATPase activity (Kuczmarski and Spudich, 1980). Phosphorylation of the heavy chain with a kinase from cells developed by starvation was also shown to inhibit actin-activated ATPase activity (Maruta et al., 1983). All heavy chain phosphorylation sites recognized by these kinases, as well as the in vivo site, have been localized to the tail portion of the molecule (Claviez et al., 1982; Peltz et al., 1981). Pagh et al. (1984) have recently shown that the sites phosphorylated by a heavy chain kinase from developed amebas map adjacent to a region of the tail that is necessary for polymerization.

Previous investigations have been directed at attempting to correlate myosin phosphorylation with chemotaxis toward cAMP (Malchow et al., 1981; Rahmsdorf et al., 1978). A protein that comigrated with the heavy chain on SDS gels and cosedimented with myosin in actomyosin precipitates was phosphorylated in Triton X-100 extracts of developed amebas. We have reproduced this result and further shown by immunoprecipitation that this in vitro phosphorylated 210,000 dalton protein is the myosin heavy chain.

Malchow et al. (1981) and Rahmsdorf et al. (1978) also showed that, after cAMP stimulation of intact cells, there was a transient increase in the rate of myosin phosphorylation that occurred in cell extracts. On the basis of these results, they proposed that the in vitro increase in myosin heavy chain phosphorylation was due to transient inhibition of the myosin heavy chain kinase in vivo. An alternative interpretation of their results is that cAMP triggers a transient increase in the amount of myosin heavy chain phosphorylation in vivo. To distinguish between these two possibilities we directly examined in vivo levels of myosin phosphorylation before and after cAMP stimulation. We demonstrate here that, in vivo, phosphorylation levels of both the heavy chain and the 18,000 dalton light chain increase after stimulation of amebas with cAMP. Thus, it is likely that the increased rate of phosphorylation in vitro reflects the increased rate of myosin phosphorylation in vivo. The kinetics of the myosin phosphorylation increases correlate with those of a cAMP-induced shape change of the intact cell.

Results

Myosin Can Be Specifically Immunoprecipitated from ^{32}P-Labeled Amebas

Figure 1, lane E, shows Coomassie stain of the Dictyostelium proteins immunoprecipitated by our polyclonal antimyosin IgG. The proteins seen in addition to the myosin heavy chain come from the antibody preparation or the Staphylococcus A cells. When myosin was immunoprecipitated from ^{32}P-labeled amebas, the only ^{32}P-labeled proteins seen in the autoradiograph of the immunoprecipitate (Figure 1, lane E′) comigrated with the heavy and light chains of myosin from Dictyostelium. Virtually all of the total myosin was immunoprecipitated by antimyosin antibodies (Figure 1, compare lanes A, D, and E), whereas no myosin was immunoprecipitated by the preimmune serum (lane C). The lysis buffer was chosen to prevent proteoly-

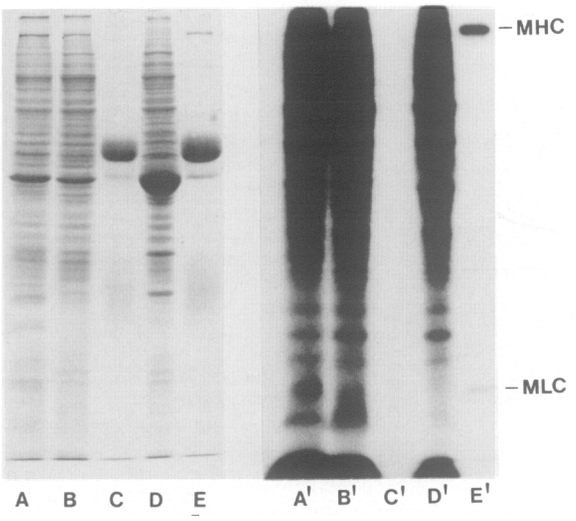


Figure 1. Specificity of Immunoprecipitation for Dictyostelium discoideum Myosin

A suspension of developed amebas was treated with 5 mM caffeine for 30 min and then labeled with [^{32}P]orthophosphate (0.5 mCi/ml) for 30 min. Aliquots of the suspension were taken, immunoprecipitated, and subjected to electrophoresis as described in Experimental Procedures. Lanes A–E, Coomassie-stained 12% polyacrylamide gel; lanes A'–E', autoradiographs of A–E; Lane A, total lysate; lane B, supernatant from 20,000 rpm centrifugation of lysate; lane C, final washed pellet after immunoprecipitation with preimmune serum; lane D, supernatant from immunoprecipitation with antimyosin antibodies; lane E, final washed pellet after immunoprecipitation with antimyosin antibodies.

sis (Figure 1, compare lane A, total lysate which was immediately TCA-precipitated, with lane E, myosin immunoprecipitate). In lysis buffer, the myosin was stable for several hours on ice, as judged by SDS–polyacrylamide gel electrophoresis. The lysis buffer also prevents phosphatase activity as determined by the following criterion. Myosin labeled with ^{32}P in vitro with a specific heavy chain kinase (Kuczmarski and Spudich, 1980) was incubated with a cell lysate (or lysis buffer as a control) for 30 min on ice and then immunoprecipitated. The same counts per minute were recovered in both cases. ATP (1 mM) was included in the lysis buffer to prevent addition of [^{32}P]-phosphate to myosin by kinases after cell lysis.

Transient cAMP-Induced Changes in In Vivo Levels of Myosin Phosphorylation

Cells were labeled with [^{32}P]orthophosphate and then stimulated with cAMP as indicated in Figure 2A. Figure 2B shows an autoradiograph of the immunoprecipitates from a typical experiment. Figure 2C shows quantitation of these results by densitometry. Light chain phosphorylation transiently increased by a factor of 3.2 after stimulation (SD = 0.4, n = 3). Heavy chain phosphorylation increased by a factor of 1.8 (SD = 0.4, n = 3). Phosphoamino acid analysis revealed that all of the light chain phosphorylation occurred on serine. Phosphoamino acid analysis of the heavy chain revealed both serine and threonine phosphorylation, with the phosphorylation increase occurring primarily only on threonine (Berlot et al., unpublished data). In unstimulated cells there was almost twice as much phosphate on the heavy chain as on the light chain; but at the phosphorylation peak after stimulation, both subunits had about the same amount of phosphate.

During a normal response to cAMP stimulation, adenylate cyclase is activated, resulting in a transient rise in intracellular cAMP. Activation of adenylate cyclase and the rise in intracellular cAMP can be blocked by caffeine. However, caffeine has no effect on chemotaxis (Brenner and Thoms, 1984). To test whether the intracellular cAMP increase was required for the myosin phosphorylation responses, amebas were pretreated with 5 mM caffeine for 30 min, then labeled with [^{32}P]orthophosphate. After 30 min, the cells were stimulated with 2×10^{-6} M cAMP (in the presence of DTT to inhibit the extracellular cAMP phosphodiesterase). The phosphorylation increases in both the heavy chain and the 18,000 dalton light chain were still observed (Figure 2D). In fact, the increases in the light chain were greater than without caffeine. In the presence of caffeine, light chain phosphorylation increased 9.8-fold (SD = 0.5, n = 5) and heavy chain phosphorylation increased 1.6-fold (SD = 0.3, n = 5) (Figures 2D and 2E). The greater increases in phosphorylation appeared to be due to a reduction in the basal phosphorylation levels rather than to an elevation of the peak phosphorylation levels (see quantitation below).

In the absence of caffeine, phosphorylation peaked for both the light chain and the heavy chain 30 sec after the stimulus. Phosphorylation returned to prestimulus levels after about 2 min (Figure 2C). Caffeine increased the rise and recovery times for the myosin phosphorylation responses (Figure 2E). The phosphorylation levels peaked at 40–60 sec and then decreased with a $t_{1/2} \simeq 3$ min. Immediately before the phosphorylation increase, the heavy chain also exhibited a transient decrease in phosphorylation. This was most clearly observed in the presence of caffeine. This pattern of myosin phosphorylation may underlie the series of shape changes induced in amebas exposed to a temporal increase in cAMP concentration. This stimulus causes amebas to cease random movement for about 20 sec, a response referred to as a "cringe" (Futrelle et al., 1982). The cells then extend pseudopods in all directions, which causes them to flatten on the substrate. After approximately 2–3 min, the amebas adapt to the stimulus and resume random movement (Fontana et al., 1985; Chisholm et al., 1985).

The following results suggest that the increased ^{32}P associated with myosin after cAMP stimulation reflects an increase in the extent of phosphorylation rather than increased turnover of phosphate. Although the specific radioactivity of the intracellular ATP pools was not equilibrated during these experiments, there was no detectable increase in the specific radioactivity of ATP during the short period in which time points were taken or when cAMP was added. There was also no increase in the specific radioactivity of myosin in the absence of the cAMP stimulus during this interval (data not shown). Furthermore, the transience of the response suggests that the cAMP-stimulated increase in myosin phosphorylation is not due to an exchange of [^{32}P]orthophosphate for unlabeled phosphate on myosin.

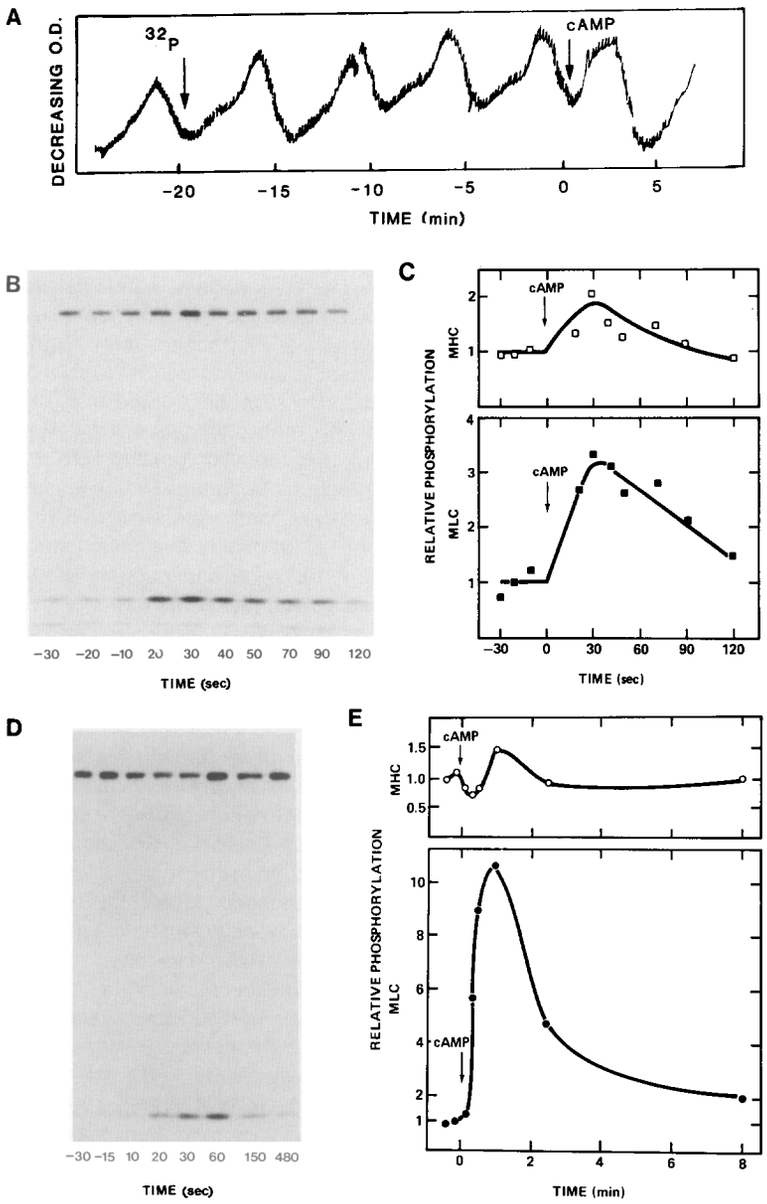


Figure 2. Transient Increase in In Vivo Levels of Myosin Phosphorylation after cAMP Stimulation

(A) Spontaneous light-scattering changes due to cAMP signaling. The developed cell suspension was labeled with [^{32}P]orthophosphate (0.1 mCi/ml) for 20 min, then stimulated with 2×10^{-6} M cAMP. At the times indicated in (B), aliquots of the suspension were taken, immunoprecipitated for myosin, and subjected to SDS–polyacrylamide gel electrophoresis on a 12% polyacrylamide gel as described in Experimental Procedures.

(B) Autoradiograph of gel of immunoprecipitated time points.

(C) Relative phosphorylation of myosin versus time after cAMP stimulation.

(D) A developed cell suspension was pretreated with caffeine for 30 min, then labeled with ^{32}P (0.5 mCi/ml) for 30 min. A stimulus of 2×10^{-6} M cAMP plus 10 mM DTT was applied and time points were taken and immunoprecipitated. The autoradiograph of the 12% polyacrylamide gel is shown.

(E) Relative phosphorylation of myosin versus time after cAMP stimulation in the presence of caffeine. Relative phosphorylation was quantitated using densitometry as described in Experimental Procedures.

Gerisch et al. (1979) determined that the intracellular ATP concentration for Dictyostelium is approximately 1 mM and does not change during spontaneous oscillations. Given this information, it was possible to compare the amounts of ^{32}P incorporated into known quantities of cellular ATP and of the heavy and light chains of myosin. From these ratios the mole fractions of incorporated phosphate to myosin subunit were computed. About 5% of myosin heavy chains and 3% of myosin light chains are phosphorylated in the unstimulated state in the absence of caffeine. In the presence of caffeine, basal light chain phosphorylation levels are reduced to a much greater extent than those of the heavy chain. Under these conditions about 3% of myosin heavy chains and 0.7% of myosin light chains are phosphorylated before cAMP stimulation. After cAMP stimulation the phosphorylation levels of the light chain are similar, with or without pretreatment with caffeine. Thus it is mainly the difference in basal phosphorylation levels that accounts for the greater increases in the presence of caffeine.

Dose Response for In Vivo Myosin Phosphorylation Increases

Since the time course of the myosin phosphorylation responses does not vary with the concentration of the cAMP stimulus (see below), it was possible to look at the relationship between cAMP dose and myosin phosphorylation response at any time during the response. We chose to compare responses near the peak of myosin light chain phosphorylation (1 min in the presence of caffeine). As seen in Figure 3, the K_{50} (the concentration of cAMP which induced a half-maximal myosin phosphorylation change) is approximately 5×10^{-9} M. This value is close to the K_{50} for chemotaxis, which is 3×10^{-9} M (Van Haastert, 1983).

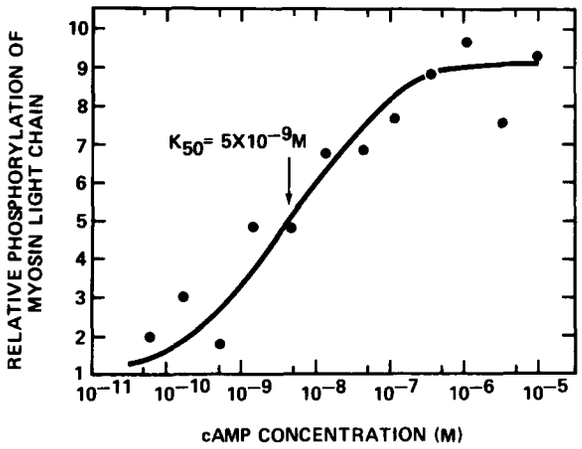


Figure 3. Dose Dependence of In Vivo Myosin Light Chain Responses to Applied cAMP Stimuli

A developed cell suspension was treated with 5 mM caffeine for 30 min, followed by [^{32}P]orthophosphate (0.5 mCi/ml) for an additional 30 min. Immediately before stimulus application, 10 mM DTT was added. Aliquots of the suspension were then added to tubes containing cAMP at the indicated concentrations and shaken for 1 min before addition of ice-cold 2× lysis buffer. Samples were immunoprecipitated for myosin and subjected to SDS–polyacrylamide gel electrophoresis. Relative phosphorylation was quantitated using densitometry.

Relative Adaptation and Additivity of In Vivo Myosin Phosphorylation Responses

The transient nature of the myosin phosphorylation increases in response to a constant cAMP stimulus implies an endogenous mechanism for terminating the responses, a phenomenon called adaptation. The adaptation process was further characterized by stimulating amebas with a constant, subsaturating amount of cAMP, followed by a stimulus of a higher concentration. We first stimulated amebas with 3×10^{-9} M cAMP for 15 min and then raised the cAMP concentration to 10^{-6} M. The magnitude of the myosin light chain phosphorylation response to the first stimulus, which was approximately at K_{50}, was lower than that seen with a saturating (10^{-6} M) stimulus, but the time course was similar (compare Figures 4A and 4B). When the stimulus was raised from 3×10^{-9} M to 10^{-6} M the light chain phosphorylation again increased with the same time course as the first response (Figure 4A). The sum of the phosphorylation responses to these two consecutive stimuli (Figure 4B, open circles) was equal to the response to a single 10^{-6} M stimulus (Figure 4B, closed circles).

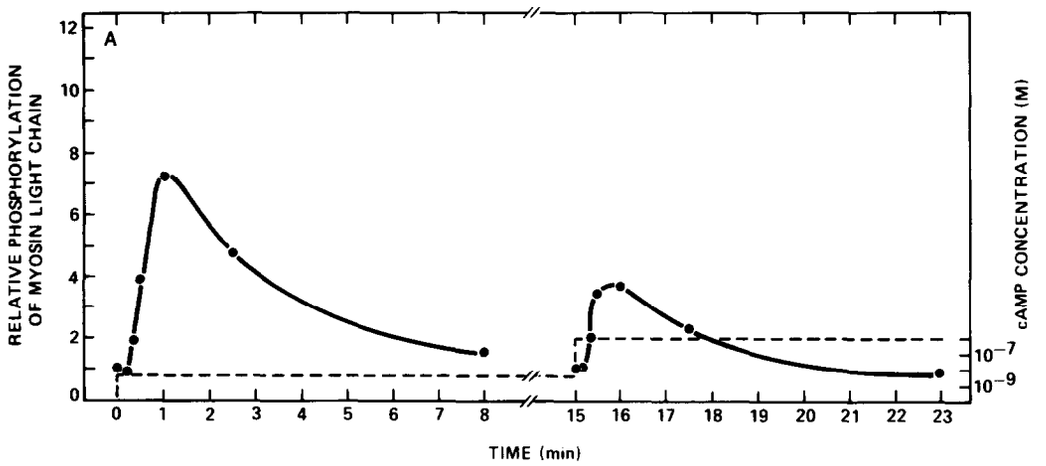
In Vitro Measurements of cAMP-Induced Changes in the Rate of Myosin Phosphorylation

When a total lysate of amebas labeled in vitro with γ-[^{32}P] ATP was analyzed by polyacrylamide gel electrophoresis, a band which comigrated with the heavy chain was by far the most heavily phosphorylated protein (Figure 5A). When myosin was immunoprecipitated, this radioactive band was selectively precipitated, indicating that it was myosin (Figure 5B). When cells were lysed into the reaction mixture, the rate of phosphorylation was constant for at least 1 min; therefore all rate measurements were taken at 30 sec. Following cAMP stimulation of intact cells, the heavy chain phosphorylation rate in vitro was found to increase transiently by a factor of 4.2 (SD = 1.1, n = 8). Addition of cAMP to lysates of unstimulated cells had no effect (data not shown). Figures 5A, 5B, and 5C (squares) show the results of a typical experiment in which heavy chain phosphorylation increased 3.3-fold.

The effects of caffeine on the in vitro measurements were similar to its effects in vivo. Consistent with the in vivo measurements of the amount of myosin phosphorylation, the peak rate of myosin phosphorylation in vitro was greater when amebas were pretreated with caffeine before cAMP stimulation. In the presence of caffeine, the increase in heavy chain phosphorylation was 9-fold (SD = 2, n = 4). Figure 5C, autoradiograph and circles, shows a typical experiment. Caffeine also affected the time course of phosphorylation. In untreated cells, the peak rate of heavy chain phosphorylation occurred at 20 and 30 sec and the rate returned to the baseline by about 50 sec after stimulus addition. In the presence of caffeine, the peak occurred at approximately 45–55 sec and was broader. Immediately before the phosphorylation increase, between 5 and 10 sec after the stimulus, there was a decrease in the rate of phosphorylation. This was most clearly observed and was longer in the presence of caffeine.

To determine the dose dependence of the in vitro myosin phosphorylation responses we measured rates at the peaks of the responses (50 sec in caffeine) since the time courses were the same at different cAMP concentrations. The K_{50} (1.2×10^{-8} M) was similar to that observed in vivo. To characterize the adaptation properties of the in vitro heavy chain phosphorylation responses we applied serial increments in cAMP concentration in the manner described above for the in vivo responses. Within 15 min of continuous stimulation, the cells adapted to 5×10^{-8} M cAMP. When the stimulus was raised to 2×10^{-6} M, a second response was obtained. The sum of the responses to the two smaller increments was equal to the response to the single larger increment (data not shown).

Under the conditions described above, light chain phosphorylation was not observed. Because many Dictyostelium proteins were found to be phosphorylated in vivo (Figure 1), while the heavy chain of myosin was the only major protein observed to be phosphorylated under the in vitro phosphorylation conditions we used, we hypothesized that the heavy chain of myosin and its kinase might be associated. Support for this idea came from the fact that the reaction mixture greatly dilutes the cells (300-fold), assuming a cell volume of 5.2×10^{-10} cm^3 (Bumann et al., 1984). Increasing the dilution still further by a factor of 7 did not decrease the incorporation of ^{32}P into the heavy chain (data not shown). This dilution experiment also suggested that the light chain and its kinase might be too dilute in vitro for an effective phosphorylation reaction. This possibility was tested by adding back excess substrate. This was done by supplementing the in vitro reactions with immunoprecipitated myosin from developed cells that had been treated with caffeine and were therefore in the basal state (see Experimental Procedures). Light chain phosphorylation appeared under these conditions. After cAMP stimulation of the cell suspension, phosphory-



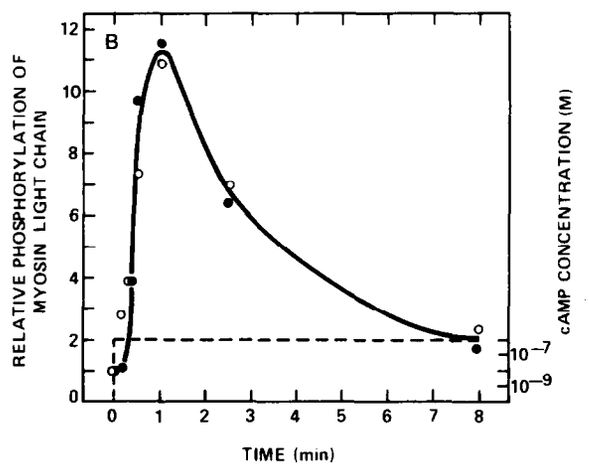


Figure 4. Relative Adaptation of In Vivo Myosin Light Chain Responses to Applied cAMP Stimuli

A developed cell suspension was treated with 5 mM caffeine for 30 min, followed by [^{32}P]orthophosphate (0.5 mCi/ml) for an additional 30 min. Immediately before stimulus application, 10 mM DTT was added. The suspension was divided into three parts. The first was stimulated with 3×10^{-9} M cAMP (first response in A). The second was first stimulated with 3×10^{-9} M cAMP during the last 15 min of the ^{32}P-labeling period and then restimulated with 10^{-6} M cAMP (second response in A). The third was stimulated with 10^{-6} M cAMP (B). Dotted line, applied stimulus. Closed circles, measured amounts of myosin phosphorylation in immunoprecipitates as determined by densitometry. Open circles (in B), sum of the two myosin phosphorylation responses in (A).

lation levels rose 18-fold (SD = 2, n = 2) and peaked approximately 30 sec after application of the stimulus (Figure 6). Heavy chain phosphorylation occurred under these conditions, but did not increase after stimulation. One possible explanation for this observation is that the heavy chain and its kinase may need to be associated in some form of complex for a stimulus-dependent phosphorylation increase to occur. In this experiment the kinase and its heavy chain substrate were added separately from different populations of cells. Minimal amounts of phosphorylated myosin contributed by the lysate appeared in a control where an unsupplemented in vitro reaction was stopped with lysis buffer and then incubated with a myosin immunoprecipitate (lane C_1, Figure 6). The immunoprecipitate did not contribute myosin kinase activity because no myosin phosphorylation occurred when the complex was incubated with γ-[^{32}P]ATP without the addition of a cell extract (lane C_2, Figure 6).

Specific radioactivity determinations of the myosin heavy chain phosphorylated in vitro and comparison with the specific radioactivity of the γ-[^{32}P]ATP in the reaction mixture revealed that, as in the case of the in vivo phosphorylations, only a small fraction of myosin molecules was phosphorylated (about 3% in the stimulated state). Phosphopeptide mapping and phosphoamino acid analysis of the heavy and light chains of myosin phosphorylated either in vivo or in vitro (Berlot et al., unpublished data) revealed that the sites phosphorylated using these two methods were very similar, the only difference being two extra chymotryptic phosphopeptides on the heavy chain phosphorylated in vivo.

Discussion

By immunoprecipitation of myosin from D. discoideum extracts phosphorylated in vitro, we have confirmed previous reports (Malchow et al., 1981; Rahmsdorf et al., 1978) which suggested a connection between myosin heavy chain phosphorylation and cell surface cAMP binding. However, our direct observations of in vivo levels of myosin phosphorylation suggest a totally different interpretation of the effect of cAMP from that put forward in these earlier reports. We observe a transient increase in phosphorylation of the heavy chain after cAMP stimulation rather than the decrease which they postulated must occur in vivo. In addition we observe a 3-fold increase in

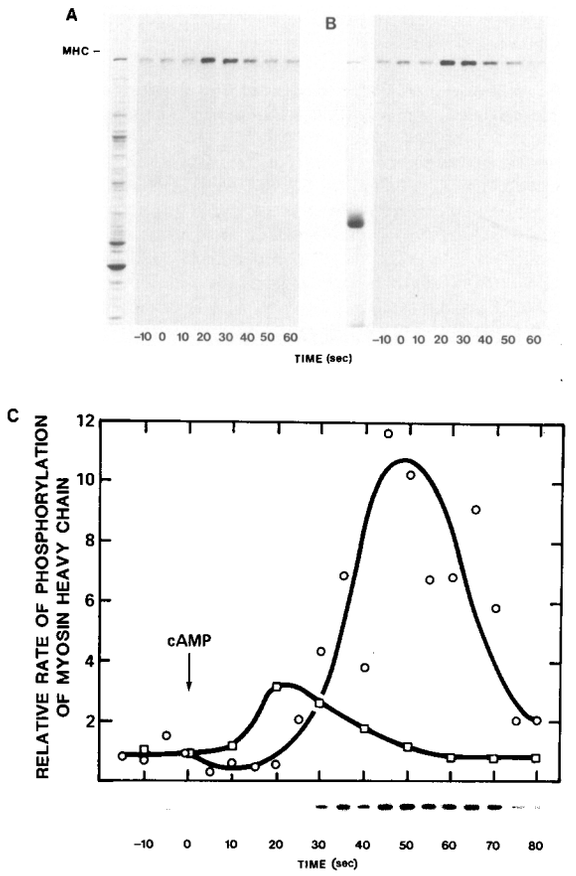


Figure 5. In Vitro Phosphorylation of the Myosin Heavy Chain by γ-[^{32}P]ATP in Triton Extracts Transiently Increases after cAMP Stimulation of Intact Amebas

The developed cell suspension was monitored for light scattering and stimulated with 2 × 10^{-6} M cAMP approximately 1 min before an anticipated endogenous signal (A, B, and squares in C) or pretreated with 5 mM caffeine for 30 min prior to stimulation (autoradiograph and circles in C). At the indicated times cells were added to the reaction mixture (see Experimental Procedures). For (A) and (B), after 30 sec at room temperature, each reaction was stopped by the addition of ice-cold 2× lysis buffer. The samples were then split in half. To one half, ice-cold TCA was added and samples were processed as described in Experimental Procedures. (A) shows an example of a well from the Coomassie-stained 7.5% polyacrylamide gel (first lane), and the autoradiograph of this gel (second–ninth lanes). The other half of the samples were immunoprecipitated for myosin. (B) shows an example of a well from the Coomassie-stained 7.5% polyacrylamide gel (first lane) and the autoradiograph of this gel (second–ninth lanes). For the autoradiograph in (C), the reactions were stopped after 30 sec with ice-cold TCA and then processed as described. (C) shows the autoradiograph of the 10% polyacrylamide gel and quantitation of relative phosphorylation versus time for the experiments shown in (A) (squares) and (C) (circles) as determined by densitometry.

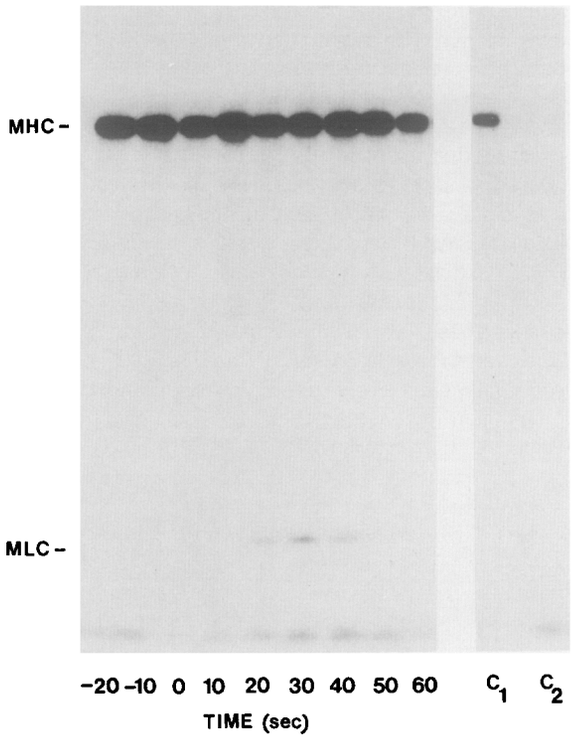


Figure 6. In Vitro Phosphorylation of the Myosin Light Chain in Triton Extracts Supplemented with Immunoprecipitated Myosin Transiently Increases after cAMP Stimulation of Intact Amebas

Developed amebas were treated with 5 mM caffeine for 30 min. The suspension was stimulated with 2 × 10^{-6} M cAMP, and at the indicated times aliquots were added to in vitro reaction mixtures, which were supplemented with myosin immunoprecipitates and processed as described in Experimental Procedures. The samples were subjected to SDS–polyacrylamide gel electrophoresis on a 12% gel, which was then autoradiographed. Lanes 1–9: experimental points taken at the indicated times. Lane C_1: unsupplemented in vitro reaction using basal cells was stopped with 2× lysis buffer, incubated with a myosin immunoprecipitate, and processed as described. Lane C_2: immunoprecipitate was incubated with reaction mixture but no added cells, stopped with 2× lysis buffer, and processed as described.

phosphorylation of the light chain which occurs with kinetics similar to those of the heavy chain phosphorylation change. These in vivo phosphorylation increases also occur when the cAMP-dependent activation of adenylate cyclase is inhibited by pretreatment of amebas with caffeine, demonstrating that an increase in intracellular cAMP concentration is not necessary for the phosphorylation increases.

The cAMP-induced myosin phosphorylation responses in the presence of caffeine are greater in magnitude and take longer than the responses in untreated cells. The greater increases appear to be due to a lowering of the basal phosphorylation levels by caffeine. Because caffeine inhibits cAMP-dependent activation of cAMP synthesis and secretion, it also inhibits oscillations of spontaneous cAMP signals and responses. During spontaneous oscillations, responses to one stimulus do not completely subside before the next stimulus appears; therefore a true basal state may never be achieved. It is possible that the effect of caffeine on the myosin phosphorylation responses may be merely to allow the cells to attain a true basal state before application of the stimulus.

Our observations suggest that the cAMP-induced myosin phosphorylation responses are part of the chemotactic sensing mechanism. First, the dose response curves are similar. Second, the kinetics of the observed myosin phosphorylation increases correlate temporally with an amebal shape change response to a constant cAMP stimulus (Fontana et al., 1985; Chisholm et al., 1985). Both the shape change response and the myosin phosphorylation response in the presence of caffeine peak approximately 1 min after application of the stimulus and then

return to the baseline more slowly with a $t_{1/2}$ of approximately 3 min. The shape change is assayed under rapid perfusion conditions in which the cells do not oscillate. Caffeine has no effect on the kinetics of the shape change under these conditions. Therefore, we compare the time course of this response with that of the myosin phosphorylation response in the presence of caffeine so that oscillations are also absent. In addition, heavy chain phosphorylation transiently decreases approximately 5–15 sec after cAMP stimulation, which coincides with the "cringe" response (Futrelle et al., 1982). A similar series of cell shape changes has been observed in polymorphonuclear leukocytes in response to F-Met-Leu-Phe (Zigmond, 1974). This chemoattractant also triggers a myosin light chain phosphorylation increase (Fechheimer and Zigmond, 1983). Stimulus-induced myosin light chain phosphorylation has also been correlated with cell shape changes in platelets (Daniel et al., 1984).

Although phosphorylation of the myosin heavy chain by specific kinases has been shown to inhibit thick filament assembly and actin-activated ATPase (Kuczmarski and Spudich, 1980), increased actin-activated ATPase and motility in the Nitella-based movement assay (Sheetz and Spudich, 1983) were observed after phosphorylation of the 18,000 dalton light chain with a kinase from growing cells (Griffith and Spudich, unpublished data). An important issue for future studies is to understand how the seemingly opposite effects of simultaneous heavy chain and light chain phosphorylation are accommodated in vivo. It is possible, for example, that in chemotaxis one population of myosin filaments is disassembled at the same time that a different population of myosin is activated to move along actin. Recent work (Yumura and Fukui, 1985) has already demonstrated a cAMP-induced alteration in myosin thick filament localization. An understanding of the in vivo roles of alterations in myosin phosphorylation may require the differential cellular localization of myosin subpopulations with varying sites and/or degrees of phosphorylation.

Experimental Procedures

Cell Culture and Development

Amebas of Dictyostelium discoideum strain Ax-3 were grown in HL-5 medium (Loomis, 1971), harvested at a density of less than 5×10^6 cells/ml, washed once in MES buffer (20 mM MES [pH 6.8], 0.2 mM $CaCl_2$, 2 mM $MgSO_4$), and resuspended to a density of 2×10^7 cells/ml to initiate development. Cells were shaken at 100 rpm at 22°C for 3.5 hr prior to being used for experiments.

Light-Scattering Technique

To monitor the spontaneous oscillations in cAMP production within the cell suspension we measured light-scattering changes that occur concurrently. When cAMP is bound to specific receptors in the cell membrane, there is a transient decrease in light scattering (Gerisch and Hess, 1974; Figure 2A) thought to be due to a combination of single cell shape changes and the transient formation of aggregates. To measure light scattering, cells were shaken in a beaker at 100 rpm and pumped at maximum speed through a Gilson peristaltic pump into a glass test tube in a Beckman Model 35 spectrophotometer and then back into the beaker. Optical density was recorded at 425 nm. The time to reach a peak of decreased light scattering is about 1 min. By measuring the peak-to-peak time, the period of spontaneous oscillations (usually about 6 min) was determined. The light-scattering changes also served as a useful indicator as to whether the cells responded to an applied stimulus.

^{32}P-Labeling and cAMP Stimulation of Dictyostelium Amebas

When cells were stimulated with cAMP during oscillations the cAMP stimulus was applied about 1 min prior to a spontaneous stimulus. To measure in vivo phosphorylation the cells were labeled with [^{32}P]orthophosphate (ICN), 0.1 mCi/ml, 20 min before application of the stimulus. In some experiments caffeine was added to the suspension to suppress spontaneous oscillations and DTT was applied with the stimulus to suppress extracellular cAMP phosphodiesterase. In these experiments 5 mM caffeine was added to the suspension 30 min prior to addition of [^{32}P]orthophosphate, 0.5 mCi/ml. After an additional 30 min the cAMP stimulus and 10 mM DTT were applied. The ability to obtain myosin phosphorylation responses in the presence of caffeine made it possible to analyze the effects of subsaturating cAMP stimuli of a defined concentration. Normally this would be impossible with cells in suspension because cAMP-dependent activation of adenylate cyclase would result in a positive feedback loop of self-stimulation.

Antibodies

Polyclonal antibodies against Dictyostelium discoideum myosin were raised in two rabbits, the sera of which were pooled. IgG was purified from the serum as follows, all manipulations being performed at 4°C or on ice. Ammonium sulfate was added to 30% saturation and the serum was centrifuged at 15,000 rpm for 20 min in a Sorvall centrifuge. The supernatant was brought to 50% saturation in ammonium sulfate and similarly centrifuged. The 50% pellet was dissolved in 5 ml of phosphate-buffered saline with azide (PBS-N_3) and dialyzed for 24 hr against two changes of 2 liters of PBS-N_3. The dialysate was chromatographed on a 2.5 × 100 cm Sephadex G150 column. The peak IgG-containing fractions were pooled, aliquoted, and stored at −80°C. IgG concentration was determined according to A_{280} of 1.5 = 1 mg/ml.

Preparation of IgG–Staphylococcus A Cell Mixture

Fifty microliters of Staphylococcus A cells (Pansorbin) that had been washed three times in immunoprecipitation (IP) buffer (see below) plus 1 mg/ml ovalbumin were added to 40 μl IgG (1 mg/ml) and incubated at 4°C on a rotator for at least 30 min prior to the addition of a cell lysate.

Preparation and Immunoprecipitation of Cell Lysates

Cell suspensions containing up to 5×10^6 cells were added to an equal volume of ice-cold 2× lysis buffer (40 mM Tris-Cl [pH 7.5], 0.2% NP40, 2 mM DTT, 10 mM EDTA, 2 mM PMSF, 2 mM TAME, 200 μM TPCK, 200 μM TLCK, 20 mM $NaHSO_3$, 100 μg/ml RNAase A [Worthington], 50 mM sodium pyrophosphate, 200 mM NaF, 2 mM ATP, and 200 mM potassium phosphate [pH 7.5]) and centrifuged for 30 min at 20,000 rpm in a Sorvall centrifuge. The supernatants were added to preadsorbed IgG–Staphylococcus A cell mixtures and incubated for 30 min at 4°C with rotation. The samples were then centrifuged in a microfuge and the pellets were resuspended in 1 ml IP buffer with 1 mg/ml ovalbumin (IP buffer is 1× lysis buffer minus RNAase A). The pellets were washed three times and then once with IP buffer minus ovalbumin. The pellets were frozen and then resuspended in SDS sample buffer and boiled for 5 min. The supernatants from a microfuge spin were then loaded on a polyacrylamide gel.

In Vitro Phosphorylation of Cell Lysates

Before and after the application of a cAMP stimulus, 100 μl aliquots of developed cells were withdrawn from a shaking suspension and added to 200 μl of a reaction mixture containing 0.2% Triton X-100, 2 mM $MgCl_2$, 7.5 mM Tris-Cl (pH 7.5), and 20 μM γ-[^{32}P]ATP (10^4 Ci/mol, from Amersham or prepared according to the method of Walseth and Johnson, 1979) and incubated for 30 sec at 22°C. Reactions were stopped by the addition of an equal volume of either ice-cold 4% TCA or 2× lysis buffer. Samples stopped with TCA were pelleted in a microfuge after 30 min on ice, washed once with 1 ml 50 mM Tris-Cl (pH 7.5), 1 mM DTT, and then resuspended in SDS sample buffer. Samples stopped with 2× lysis buffer were immunoprecipitated as described.

Preparation of Immunoprecipitate for Subsequent In Vitro Phosphorylation

Myosin was immunoprecipitated from developed cells that had been

treated with caffeine and were therefore in the basal state. An excess amount of lysate (from 2 × 10^7 cells) was added to each Staphylococcus A cell–antibody complex to saturate the vast majority of available antigen binding sites on the antibody. Aliquots of the myosin-antibody–Staphylococcus A cell complex were added to the in vitro reaction mixture, and the in vitro reactions were performed on lysates from cell aliquots taken from a suspension before and at 10 sec intervals after cAMP stimulation. The amount of myosin added to the reactions in the form of immunoprecipitates was approximately 10-fold the amount in the kinase-containing cell aliquots. The reactions were stopped by adding an equal volume of 2× lysis buffer and the immunoprecipitates were washed again as if another immunoprecipitation were being performed.

Polyacrylamide Gel Electrophoresis

Polyacrylamide gel electrophoresis was performed according to Laemmli (1970). The gels were stained with Coomassie Blue, dried on a Hoefer slab gel drying apparatus (Hoefer Scientific Instruments, San Francisco, California), and exposed to Kodak XAR-5 film with Cronex lightning plus intensifying screens (Dupont) at −80°C.

Densitometry

Gels and autoradiographs were scanned with an RFT Scanning Densitometer (Transidyne General Corporation, Ann Arbor, Michigan) at 600 nm and the peaks were cut out and weighed. Relative phosphorylation of myosin was determined by dividing the weights of the heavy chain or light chain autoradiograph peaks by the weights of the corresponding heavy chain Coomassie peaks.

Determination of Specific Radioactivity of Cellular ATP Pools

ATP specific radioactivity was measured by adding aliquots of ^{32}P-labeled cells to an equal volume of 0.1 N perchloric acid, precipitating out the inorganic phosphate as described (Sugino and Miyoshi, 1964), loading the supernatant with an ATP standard on PEI cellulose (Polygram CEL 300 PEI/UV_{254}, Macherey-Nagel), and chromatographing in 0.5 M ammonium sulfate for the first third of the plate followed by 0.7 M ammonium sulfate. ATP spots were localized by visualization of the standards with UV light. The spots were cut out and Cerenkov radiation was measured. Recovery of cellular ATP by this method was 75%, as determined by the use of internal standards.

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